Model Question Paper-1 with effect from 2022-23 (CBCS Scheme)

USN | | | | | | | | | | |

Fourth Semester B.E. Degree Examination

Subject Title Principles of Communication Systems

TIME: 03 Hours **Max. Marks: 100**

Note: 01. Answer any **FIVE** full questions, choosing at least **ONE** question from each **MODULE**.
02.
03.

		Module -1	*Bloom's Taxonomy Level	Marks
Q.01	a	Define Probability. Illustrate the relationship between sample space, events and probability.	L2, CO5	06
	b	Define the autocorrelation and cross-relation functions. Infer the properties of autocorrelation function.	L2, CO5	06
	c	Develop a program to generate the probability density function of Gaussian distribution function.	L3, CO5	08
		OR		
Q.02	a	What is conditional probability? Prove that P(B/A) = P(A/B).P(B)/P(A)	L2, CO5	06
	b	Outline random processes and illustrate an ensemble of sample functions with a neat diagram.	L2, CO5	06
	c	Show that, if a Gaussian process X(t) is applied to a stable linear filter, then the random process Y(t) developed at the output of the filter is also Gaussian.	L3, CO5	08
		Module-2		
Q. 03	a	Interpret the concepts of modulation index and percentage of modulation. Write the necessary equations.	L2, CO1	08
	b	A standard AM broadcast station is allowed to transmit modulating frequencies upto 5 KHz. If the AM station is transmitting on a frequency of 980 KHz, compute the maximum and minimum upper and lower sidebands and the total bandwidth occupied by the AM station.	L3, CO1	05
	c	Explain high-level collector modulator with a neat block diagram.	L2, CO1	07
		OR		
Q.04	a	Outline a diode detector AMD modulator with necessary block diagram and waveforms.	L2, CO1	08
	b	An AM transmitter has a carrier power of 30W. The percentage of modulation is 85 percent. Calculate (i) the total power and (ii) the power in one sideband.	L3, CO1	05
	c	Explain a general block diagram of an FDM system.	L2, CO1	07
		Module-3		
Q. 05	a	Compare and contrast FM and AM.	L3, CO2	06
	b	Explore with a neat diagram the concept of frequency modulation with an IC VCO.	L2, CO2	07
	c	Draw the block diagram of a super heterodyne receiver and explain the function of each block.	L2, CO2	07
		OR		
Q. 06	a	Identify a method used to convert a phase-modulated (PM) signal into a frequency-modulated (FM) signal.	L3, CO2	06
	b	Define PLL. Explain the basic block diagram of a PLL along with capture and lock ranges.	L2, CO2	07
	c	Interpret the concept of a mixer with a neat schematic diagram.	L2, CO2	07
		Module-4		
Q. 07	a	What are the advantages of digital signals over analog signals?	L1, CO3	04

	b	State sampling theorem. Explain sampling with neat sketches and equations. What are the challenges faced with Nyquist criteria for sampling? Develop a program to display the signals and its spectrum.	L3, CO3	10
	c	Explain the generation and detection of PPM waves with a relevant block diagram.	L2, CO3	06
		OR		
Q. 08	a	What is aperture effect in PAM systems? How can it be minimized?	L1, CO3	04
	b	What is multiplexing and why is it required in communication? Explain the working of TDM with a neat block diagram.	L3, CO3	10
	c	Explain the basic elements of a PCM system with neat diagrams.	L3, CO3	06
		Module-5		
Q. 09	a	Define Intersymbol Interference (ISI). Outline Baseband binary data transmission system with neat block diagram and equations.	L2, CO4	08
	b	Develop a code to generate and plot eye diagram.	L3, CO4	06
	c	Illustrate the concept of noise in cascaded stages with a diagram. Write Friis' formula and mention its terms.	L2, CO1	06
		OR		
Q. 10	a	Explain the following concepts briefly: (i) Nyquist criterion for distortionless transmission. (ii) Baseband M-ary PAM transmission	L2, CO4	08
	b	Develop a code to generate NRZ and RZ pulse.	L3, CO4	06
	c	Define Signal-to-Noise Ratio (SNR). Explain the different types of external and internal noise.	L2, CO1	06

*Bloom's Taxonomy Level: Indicate as L1, L2, L3, L4, etc. It is also desirable to indicate the COs and POs to be attained by every bit of questions.

Model Question Paper-2 with effect from 2022-23 (CBCS Scheme)

USN

Fourth Semester B.E. Degree Examination

Subject Title: Principles of Communication Systems

TIME: 03 Hours **Max. Marks: 100**

Note: 01. Answer any **FIVE** full questions, choosing at least **ONE** question from each **MODULE**.
02.
03.

		Module -1	***Bloom's Taxonomy Level**	**Marks**
Q.01	a	A random variable is said to be uniformly distributes over the interval (a,b). Determine the probability density function of uniformly distributed RV.	L2, CO5	06
	b	Determine the characteristic function of a Gaussian random variable with a given mean and variance.	L2, CO5	08
	c	Prove the following two properties of the Autocorrelation Function of a Random Process i)If X(t) contains a dc component equal to A, then Rx(t) will contain a constant amplitude equal to A^2. ii) If X(t) contains a sinusoidal component, then Rx(t) will also contain a sinusoidal component of the same frequency	L2, CO5	06
		OR		
Q.02	a	Define correlation, covariance and correlation coefficient	L2, CO5	04
	b	Explain central limit theorem as applied to Gaussian Random Process	L2, CO5	06
	c	Define Autocorrelation Function and Crosscorrelation Function State and prove the properties of ACF	L3, CO5	08
		Module-2		
Q. 03	a	Explain amplitude modulation in time domain and frequency domain with necessary expressions and illustrations.	L2, CO1	08
	b	With necessary schematic of square -law circuit and expressions, explain the generation of AM.	L3, CO1	05
	c	Write a MATLAB code to generate Amplitude Modulation and demodulation waveforms and display its spectrums.	L2, CO1	07
		OR		
Q.04	a	An AM transmitter uses high-level modulation of the final RF power amplifier. Which has a de supply voltage Vcc of 48 V with a total current I of 3.5 A. The efficiency is 70 percent. a. What is the RF input power to the final stage? b. How much AF power is required for 100 percent modulation?	L2, CO1	08
	b	Explain with neat diagrams, amplitude demodulator using the diode detector	L3, CO1	05
	c	Explain with diagrams, the working principle of Lattice-type balanced modulator.	L2, CO1	07
		Module-3		
Q. 05	a	The input to an FM receiver having an S/N of 2.8. The modulating frequency is 1.5 kHz. The maximum permitted deviation is 4 kHz. What are (a) the frequency deviation caused by the noise and (b) the improved output SIN?	L3, CO2	06
	b	Explain with a neat diagram, the frequency spectrum of FM Modulated wave	L2, CO2	07
	c	Explain with diagrams, the working principle of Frequency-Modulation using Crystal Oscillator	L2, CO2	07
		OR		
Q. 06	a	Identify the Noise Suppression Effects of FM.	L3, CO2	06

	b	Compare and contrast FM using crystal oscillator circuits with FM using varactors	L2, CO2	07
	c	Explain general block diagram of a super-heterodyne receiver.	L2, CO2	07
		Module-4		
Q. 07	a	State and prove sampling theorem	L1, CO3	04
	b	For the data stream [01101001], Draw the following line code waveforms i)Unipolar NRZ ii)Polar NRZ iii)Unipolar RZ iv)Bipolar RZ v)Manchestor code	L3, CO3	10
	c	Explain the generation and detection of PPM waves with a relevant block diagram.	L2, CO3	06
		OR		
Q. 08	a	What is aperture effect in PAM systems? How can it be minimized?	L1, CO3	04
	b	What is multiplexing and why is it required in communication? Explain the working of TDM with a neat block diagram.	L3, CO3	10
	c	Exercise examples from Text 2 in the prescribed syllabus, 7.3, 7.4, 7.14,7.17	L3, CO3	06
		Module-5		
Q. 09	a	Define Intersymbol Interference (ISI). Outline Baseband binary data transmission system with neat block diagram and equations.	L2, CO4	08
	b	Develop a code to generate and plot eye diagram.	L3, CO4	06
	c	Explain bandwidth requirements of T1 system	L2, CO1	06
		OR		
Q. 10	a	Explain the following concepts briefly: (i) Nyquist criterion for distortionless transmission. (ii) Baseband M-ary PAM transmission	L2, CO4	08
	b	Develop a code to generate NRZ and RZ pulse.	L3, CO4	06
	c	Define Signal-to-Noise Ratio (SNR). Explain the different types of external and internal noise.	L2, CO1	06

*Bloom's Taxonomy Level: Indicate as L1, L2, L3, L4, etc. It is also desirable to indicate the COs and POs to be attained by every bit of questions.

CBCS SCHEME

USN | | | | | | | | | |

BEC402

Fourth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Principles of Communication Systems

Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	M	L	C
Q.1	a.	Define probability. Illustrate the relationship between sample space, events and probability.	06	L1	CO1
	b.	Outline random processes and illustrate an ensemble of sample function with a neat diagram.	06	L2	CO1
	c.	Show that if a Gaussian process x(t) is applied to a stable linear filter, then the random process y(t) developed at the output of the filter is also Gaussian.	08	L3	CO2
		OR			
Q.2	a.	What is conditional probability? Prove that $P(B/A) = P(A/B) \cdot P(B) / P(A)$	06	L1	CO1
	b.	Define mean, correlation and covariance function.	06	L2	CO2
	c.	Develop a program to generate the probability density function of Gaussian distribution function.	08	L3	CO2
		Module – 2			
Q.3	a.	An antenna has an impedance of 40 Ω an unmodulated AM signal produces a current of 4.8 A. The modulation is 90 percent calculate i) The carrier power ii) The total power iii) The sideband power	06	L1	CO1
	b.	Explain with neat diagrams amplitude demodulation using the diode detector.	07	L1	CO1
	c.	Explain a general block diagram of an FDM system	07	L2	CO2
		OR			
Q.4	a.	Interpret the concept of modulation index and percentage of modulation write the necessary equations.	06	L1	CO1
	b.	Explain high level collector modulation with neat block diagram.	07	L2	CO1
	c.	Explain with diagrams the working principle of lattice type balanced modulator.	07	L2	CO2
		Module – 3			
Q.5	a.	Compare and contrast FM and AM.	06	L1	CO1
	b.	Explain with diagrams the working principle of frequency modulation using voltage controlled oscillator.	07	L2	CO2
	c.	Explain general block diagram of a super heterodyne receiver.	07	L2	CO2
		OR			
Q.6	a.	The input to an FM receiver having an S/N of 2.8. The modulating frequency is 1.5 KHz. The maximum permitted deviation is 4 KHz. What are (i) The frequency deviation caused by the noise (ii) The improved output S/N.	06	L2	CO2
	b.	Define PLL. Explain the basic block diagram of a PLL.	07	L1	CO2
	c.	Explain JFET mixer.	07	L2	CO2

		Module – 4			
Q.7	a.	What are the advantages of digital signal over analog signals?	04	L1	CO1
	b.	Explain with basic elements of a PCM system with neat diagrams.	08	L2	CO1
	c.	For the data stream 0 1 1 0 1 0 0 1 draw the following line code waveforms i) Unipolar NRZ ii) Polar NRZ iii) Unipolar RZ iv) Manchestor code	08	L3	CO2
		OR			
Q.8	a.	State and prove Sampling theorem.	04	L1	CO1
	b.	What is multiplexing and why is it required in communication? Explain the working of TDM with a neat block diagram.	08	L2	CO1
	c.	Explain the generation of PPM with a relevant block diagrams and waveforms.	08	L2	CO2
		Module – 5			
Q.9	a.	Define Intersymbol interference (ISI) outline baseband binary data transmission system with neat block diagram and equations.	08	L2	CO1
	b.	Develop a code to generate RZ pulse.	04	L3	CO2
	c.	Define signal to noise ratio. Explain different types of external and internal noise.	08	L2	CO1
		OR			
Q.10	a.	Explain the following concept briefly: i) Nyquist criterion for distributors transmission ii) Baseband M-ary PAM transmission	08	L1	CO2
	b.	Develop a code to generate Raised cosine pulse.	04	L2	CO2
	c.	Illustrate the concept of noise in cascaded stages with a diagram. Write Friis formula and mention its terms.	08	L2	CO3

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CBCS SCHEME

USN

BEC402

Fourth Semester B.E./B.Tech. Degree Examination, June/July 2024
Principles of Communication Systems

Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	M	L	C
Q.1	a.	What is conditional probability? Prove that $P\left(B/A\right) = P\left(A/B\right)\cdot P(B)/P(A)$	05	L2	CO1
	b.	Define the autocorrelation and cross correlation. Discuss the properties of autocorrelation.	10	L2	CO1
	c.	Develop a program to generate the probability density function of Gaussian distribution function.	05	L3	CO1
		OR			
Q.2	a.	Define auto-covariance, random variable, cumulative distribution function and probability distribution function.	08	L1	CO1
	b.	The random variable its plot is given as $f_x(x) = 2.e^{-2x}$ for $x \geq 0$. Find the probability that it will take value between 1 and 3.	04	L3	CO1
	c.	Define probability with an example. Discuss their properties (axioms).	08	L2	CO1
		Module – 2			
Q.3	a.	Explain amplitude modulation with necessary equations and sketches in time domain and frequency domain.	08	L3	CO2
	b.	Define modulation index and percentage of modulation. Explain over modulation and distortion.	06	L2	CO2
	c.	Derive the expression for Amplitude Modulation (AM) power in terms of modulation index.	06	L2	CO1
		OR			
Q.4	a.	Explain a general block diagram of a frequency division multiplexing.	06	L1	CO2
	b.	Explain the working principle of lattice type balanced modulator with circuit diagram.	07	L1	CO2
	c.	With neat diagrams, explain high level collector modulator.	07	L2	CO2
		Module – 3			
Q.5	a.	With a neat block diagram, explain converting a phase modulated signal into a frequency modulated signal.	07	L1	CO3
	b.	Determine the frequency modulated signal $v_{FM} = V_C \sin(2\pi f_c t + m_f \sin 2\pi f_m t)$ interms of Bessel functions. Write the amplitude of sideband frequencies (J_n) interms of modulation index (m_f).	06	L3	CO3
	c.	Identify the noise suppression of frequency modulated signal.	07	L2	CO3
		OR			
Q.6	a.	What is the maximum bandwidth of an FM signal with a deviation of 30 kHz and a maximum modulating signal of 5 kHz. (i) Using number of sidebands N = 9 (ii) Using Carson's rule	04	L2	CO3
	b.	Define phase locked loop. Explain with neat circuit diagram of FM demodulator using the IC 565.	08	L2	CO3
	c.	With neat block diagram, explain the concept of frequency modulation with an IC voltage controlled oscillator (IC NE566)	08	L2	CO3

		Module – 4			
Q.7	**a.**	Why digitize the analog signals? Explain the different processes used to convert the analog signal to digital signal.	**06**	L2	CO4
	b.	What is quantization process? Explain the different types of quantization with their important characteristics.	**07**	L2	CO4
	c.	Explain the concept of Time division multiplexing with a neat block diagram.	**07**	L2	CO4
		OR			
Q.8	**a.**	Define PCM (Pulse Code Modulation). Explain the basic elements of a PCM system with neat diagrams.	**06**	L2	CO4
	b.	For the data stream 01101001. Draw the following line code waveforms: (i) Unipolar NRZ (ii) Polar NRZ (iii) Unipolar RZ (iv) Bipolar RZ (v) Manchester code (vi) Differential coding	**09**	L3	CO4
	c.	State and prove the sampling theorem. Explain with neat sketches and equations.	**05**	L2	CO4
		Module – 5			
Q.9	**a.**	Develop a code to generate and plot eye diagram.	**06**	L3	CO5
	b.	Define noise factor and noise figure. Also explain noise in cascade connection.	**06**	L2	CO5
	c.	Define Inter Symbol Interference (ISI). Outline baseband binary data transmission system with neat block diagram and equations.	**08**	L1	CO5
		OR			
Q.10	**a.**	Explain bandwidth requirements of TI systems.	**06**	L1	CO5
	b.	Write short notes on: (i) Signal to noise ratio (ii) External noise (iii) Internal noise	**08**	L1	CO5
	c.	An RF amplifier has an S/N ratio of 8 at the input and an S/N ratio of 6 at the output. What are the noise factor, noise figure and noise temperature?	**06**	L3	CO5

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	c.	Interpret with a neat circuit diagram, the working principle of frequency modulation of a crystal oscillator with a Voltage Variable Capacitor (VVC).	8	L2	CO2
		OR			
Q.6	a.	Define Modulation. Identify any five differences between Frequency Modulation and Amplitude Modulation.	6	L2	CO2
	b.	Why Pre-emphasis and de-amphasis are required? Explain how they are implemented?	6	L2	CO2
	c.	Draw the block diagram of a super heterodyne receiver and explain the function of each.	8	L2	CO2
		Module – 4			
Q.7	a.	State and prove sampling theorem. Write a program for sampling and reconstruction of low pass signals and display the signals and its spectrum.	10	L3	CO3
	b.	Infer the working of TDM system with a neat block diagram.	5	L2	CO3
	c.	Explain briefly the block diagram of PPM generator.	5	L2	CO3
		OR			
Q.8	a.	Identify and explain the basic elements of a PCM system with neat diagrams. For the data stream [0 1 1 0 1 0 0 1], draw the following line code waveforms : (i) Unipolar NRZ (ii) Polar NRZ (iii) Unipolar RZ (iv) Bipolar RZ (v) Manchestor code	10	L3	CO3
	b.	Infer the advantages of digital signals over analog signals.	5	L2	CO3
	c.	Explain briefly the midtread and midrise Quantizers with relevant figures.	5	L2	CO3
		Module – 5			
Q.9	a.	What is Intersymbol Interference (ISI)? With a neat block diagram outline the baseband binary data transmission system and write the necessary equations?	8	L2	CO4
	b.	Define SNR. Summarize the different types of external and internal noise.	7	L2	CO4
	c.	Illustrate the concept of Noise in cascaded stages with a diagram. Write Friis formula and mention its terms.	5	L2	CO4
		OR			
Q.10	a.	What is Baseband digital transmission? Explain the following concepts briefly : (i) Nyquist criterion for distortionless transmission. (ii) Baseband M-ary PAM transmission.	8	L2	CO4
	b.	Define Noise. Classify the different types of semiconductor noise.	7	L2	CO4
	c.	What is Noise Factor and Noise Figure? An RF amplifier has an S/N ratio of 8 at the input and an S/N ratio of 6 at the output. Calculate the Noise factor and Noise figure.	5	L2	CO4

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CBCS SCHEME

USN | | | | | | | | | | |

BEC402

Fourth Semester B.E./B.Tech. Degree Examination, June/July 2025

Principles of Communication Systems

Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	M	L	C
Q.1	a.	Define Probability. Illustrate the relationship between sample space, events and probability.	6	L2	CO5
	b.	What are moments? Determine the characteristic function of a Gaussian random variable with a given mean and variance.	6	L2	CO5
	c.	Analyze the Gaussian process with Gaussian distribution curve. Infer the properties of a Gaussian process.	8	L2	CO5
		OR			
Q.2	a.	Define a random process. Interpret mean and covariance function with respect to stationary random process.	6	L2	CO5
	b.	What is Autocorrelation function? State and prove the properties of Autocorrelation function.	6	L2	CO5
	c.	Analyze the PDF and CDF of a random experiment in which three coins are tossed and condition to get random variable is getting head.	8	L3	CO5
		Module – 2			
Q.3	a.	Define Amplitude modulation. Derive an expression for Amplitude Modulation in time domain with necessary waveforms.	8	L2	CO1
	b.	A standard AM broadcast station is allowed to transmit modulating frequencies upto 5 kHz. If the AM station is transmitting on a frequency of 980 kHz, compute the maximum and minimum upper and lower side bands and the total bandwidth occupied by the AM station.	5	L3	CO1
	c.	Outline the block diagram of FDM transmitter. List the applications of FDM.	7	L2	CO1
		OR			
Q.4	a.	Develop a code to generate Amplitude Modulation Waveforms and display its spectrum.	8	L3	CO1
	b.	Apply the concept of side bands to explain DSB and SSB, draw the relevant waveforms.	5	L2	CO1
	c.	Explain with diagrams, the working principle of Lattice-type balanced modulator.	7	L2	CO1
		Module – 3			
Q.5	a.	Identify a method used to convert a Phase Modulated (PM) signal into a Frequency-Modulated (FM) signal.	6	L2	CO3
	b.	The input to an FM receives has S/N of 2.8. The modulating frequency is 1.5 kHz. The maximum permitted deviation is 4 kHz. Determine (i) The frequency deviation caused by the noise and (ii) The improved output S/N.	6	L3	CO2

CBCS SCHEME - Make-Up Exam

USN | | | | | | | | | | |

BEC402

Fourth Semester B.E/B.Tech. Degree Examination, June/July 2025
Principles of Communication Systems

Time: 3 hrs. Max. Marks: 100

Note: ***1. Answer any FIVE full questions, choosing ONE full question from each module.***
2. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	**M**	**L**	**C**
1	a.	Define the Auto correlation and Cross correlation. Discuss the properties of auto correlation.	**10**	**L2**	**CO5**
	b.	Define Probability. Illustrate the relationship between sample space , event and probability.	**5**	**L2**	**CO5**
	c.	Develop a program to generate the probability density function of Gaussian distribution function.	**5**	**L3**	**CO5**
		OR			
2	a.	Explain Conditional Probability. Prove that P(B/A) = P(A/B) . P(B) / P(A).	**6**	**L2**	**CO5**
	b.	Explain Central Limit Theorem as applied to Gaussian Random Process.	**6**	**L2**	**CO5**
	c.	Explain properties of Gaussian Process with necessary equations.	**8**	**L3**	**CO5**
		Module – 2			
3	a.	Write a MATLAB code to generate Amplitude Modulation and demodulation waveforms and display its spectrums.	**8**	**L3**	**CO2**
	b.	Explain the working principle of lattice type balanced modulator with neat circuit diagram.	**8**	**L2**	**CO2**
	c.	Ab AM transmitter has a carrier power of 30 W. The % of modulation is 85%. Calculate a) P_T b) P_{SB} in one side band	**4**	**L3**	**CO2**
		OR			
4	a.	Explain how amplitude modulated wave generated using diode modulator.	**8**	**L2**	**CO2**
	b.	Explain the working of transmitter and receiver of Frequency Division Multiplexing (FDM).	**8**	**L2**	**CO2**
	c.	Explain disadvantages of DSB and SSB.	**4**	**L2**	**CO2**

Module – 3					
5	a.	Define PLL. Explain with neat circuit diagram of FM demodulator using IC 565.	**8**	**L2**	**CO2**
	b.	Explain the demodulation process of frequency modulator using slope detector.	**8**	**L2**	**CO2**
	c.	What is the maximum Band width of an FM signal with a deviation of 30 KHz and a maximum modulating signal of 5KHz. Determine i) M.I ii) Bandwidth.	**4**	**L3**	**CO2**
OR					
6	a.	Draw the block diagram of a super heterodyne receiver and explain the function of each block.	**8**	**L2**	**CO2**
	b.	Explain the Noise suppression effects of FM with necessary waveform.	**6**	**L2**	**CO2**
	c.	Compare Amplitude Modulation (AM) versus Frequency Modulation (FM).	**6**	**L2**	**CO2**
Module – 4					
7	a.	What is quantization process? Derive output signal to noise ratio of uniform quantizer $(SNR)_0 = \left(\frac{3P}{m_{max}}\right) . 2^{2R}$ Where P average power of m(t).	**10**	**L3**	**CO3**
	b.	Explain the Regeneration of PCM waves with a neat block diagram.	**5**	**L2**	**CO3**
	c.	Explain important advantages of digital signals over analog signals.	**5**	**L2**	**CO3**
OR					
8	a.	Explain the generation and detection of PPM wave with neat diagram.	**10**	**L2**	**CO3**
	b.	An analog signal is specified by the equations : i) $x(t) = 3\cos(50\pi t) + 10\sin(300\pi t) + \cos(100\pi t)$ ii) $x(t) = \frac{1}{2\pi}\cos(4000\pi t) . \cos(1000\pi t)$.	**10**	**L3**	**CO3**
Module – 5					
9	a.	Define ISI. Explain base band binary data transmission with beat block diagram and necessary equations.	**10**	**L2**	**CO4**
	b.	Explain the different types of external and internal noise.	**5**	**L2**	**CO4**
	c.	Write a note on Eye diagram.	**5**	**L2**	**CO4**
OR					

10	a.	Explain the following concept briefly : i) Nyquist criterian for distrotionless transmission ii) Base band M – array PAM transmission iii) Band width requirement of T_1 system.	**12**	**L2**	**CO4**
	b.	With a neat diagram, explain the concept of noise in cascaded stages. Write Friis formula and mention its terms.	**8**	**L2**	**CO4**

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MODULE-2

Passing Module

1. Explain amplitude modulation with necessary equations and sketches in time domain and frequency domain.

2. Define modulation index and percentage of modulation. Explain over modulation and distortion.

3. Derive the expression for Amplitude Modulation (AM) power in terms of modulation index.

4. Explain a general block diagram of a frequency division multiplexing(FDM)

5. Explain the working principle of lattice type balanced modulator with circuit diagram.

6. With neat diagrams, explain high level collector modulator.

7. A standard AM broadcast station is allowed to transmit modulating frequencies upto 5 KHz. If the AM station is transmitting on a frequency of 980 KHz, compute the maximum and minimum upper and lower sidebands and the total bandwidth occupied by the AM station.

8. Outline a diode detector AMD modulator with necessary block diagram and waveforms.

9. An AM transmitter has a carrier power of 30W. The percentage of modulation is 85 percent. Calculate (i) the total power and (ii) the power in one sideband.

10. With necessary schematic of square -law circuit and expressions, explain the generation of AM.
Write a MATLAB code to generate Amplitude Modulation and demodulation waveforms and display its spectrums.

MODULE-4

Passing Module

1. Why digitize the analog signals? Explain the different processes used to convert the analog signal to digital signal.

2. What is quantization process? Explain the different types of quantization with their important characteristics.

3. Explain the concept of Time division multiplexing with a neat block diagram.

4. Define PCM (Pulse Code Modulation). Explain the basic elements of a
PCM system with neat diagrams.

5. For the data stream 01101001. Draw the following line code waveforms:

(i) Unipolar NRZ (ii) Polar NRZ (iii) Unipolar RZ
(iv) Bipolar RZ (v) Manchester code
(vi) Differential coding

6. What are the advantages of digital signals over analog signals?

7. What is aperture effect in PAM systems? How can it be minimized?

8. What is multiplexing and why is it required in communication? Explain the working of TDM with a neat block diagram.

9. Explain the basic elements of a PCM system with neat diagrams.

10. State sampling theorem. Explain sampling with neat sketches and equations. What are the challenges faced with Nyquist criteria for sampling? Develop a program to display the signals and its spectrum.

MODULE-5

Passing Module

1. Develop a code to generate and plot eye diagram.

2. Define noise factor and noise figure. Also explain noise in cascade connection.

3. Define Inter Symbol Interference (ISI). Outline baseband binary data transmission system with neat block diagram and equations.

4. Explain bandwidth requirements of TI systems.

5. Write short notes on:
(i) Signal to noise ratio
(ii) External noise
(iii) Internal noise

6. Illustrate the concept of noise in cascaded stages with a diagram. Write Friis' formula and mention its terms.

7. Explain the following concepts briefly:
(i) Nyquist criterion for distortionless transmission.
(ii) Baseband M-ary PAM transmission

8. Develop a code to generate NRZ and RZ pulse.

9. Develop a code to generate Raised cosine pulse.

MODULE-3

1. With a neat block diagram, explain converting a phase modulated sign into a frequency modulated signal.

2. Define phase locked loop. Explain with neat circuit diagram of FM demodulator using the IC 565.

3. With neat block diagram, explain the concept of frequency modulation with an IC voltage controlled oscillator (IC NE566).

4.Compare and contrast FM and AM.

5. Explore with a neat diagram the concept of frequency modulation with an IC VCO.

6. Draw the block diagram of a super heterodyne receiver and explain the function of each block.

7. Identify a method used to convert a phase-modulated (PM) signal into a frequency-modulated (FM) signal.

8. Define PLL. Explain the basic block diagram of a PLL along with capture and lock ranges.

9. Interpret the concept of a mixer with a neat schematic diagram.

10. Explain with diagrams, the working principle of Frequency-Modulation using Crystal Oscillator